

The Markov traces and the Fourier transforms

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Received 10 March 2005

Available online 8 November 2005

Communicated by Gunter Malle

1. Introduction

In [8], Jones has constructed a knot invariant by using a trace function on the Iwahori–Hecke algebra of type A , which is called the Markov trace. In [4], Geck and Lambropoulou have extended this theory to the case of type B . They have given the Markov trace on the Iwahori–Hecke algebra of type B and by using this, they have constructed an invariant of mixed links. They have also given the Markov trace of type D by using the fact that the Iwahori–Hecke algebra of type D can be embedded in some specialized Iwahori–Hecke algebra of type B . In this paper, we would like to consider the Markov trace on the Iwahori–Hecke algebra of any finite type.

Let $\mathcal{H}$ be an Iwahori–Hecke algebra and W the corresponding Coxeter group. It is known that the generic degrees of $\mathcal{H}$ are obtained from the fake degrees of W by applying the so-called Fourier transform which was constructed by Lusztig. The generic degrees appear in the weight of the canonical symmetrizing trace of $\mathcal{H}$. The Markov trace of $\mathcal{H}$ of type A has a parameter z and if we specialize $z \mapsto 0$, then the Markov trace specializes to the canonical symmetrizing trace of $\mathcal{H}$. So the weight of the Markov trace can be considered as an analogue of the generic degree. On the other hand, the fake degrees of W appear in the Molien series of $S(V)$, the symmetric algebra of V where V is the vector space which affords the reflection representation of W . As an analogue of the fake degree, we consider the Molien series of $S(V) \otimes L(V)$ where $L(V)$ is the exterior algebra of V . Then we determine a trace function τ on $\mathcal{H}$ whose weights are given by applying the Fourier transform to the Molien series above. When $\mathcal{H}$ is of type A , τ coincides with the Markov

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trace and when $\mathcal{H}$ is of type B or D , τ coincides with some specialized Markov trace. For any finite type, τ satisfies certain properties which allow us to call it the Markov trace on $\mathcal{H}$ (see Theorem 4.1). That is to say, the Fourier transform sends the Molien series of $S(V) \otimes L(V)$ to the weights of the Markov trace.

This paper is organized as follows. In Section 2, we calculate the Molien series of $S(V) \otimes L(V)$ in each case. In Section 3, we give the Markov trace of type A and of type B , following [8] and [4]. Based on the common property of both Markov traces, we define the Markov property (see Definition 3.7) of a trace function on any type of an Iwahori–Hecke algebra. Then we give the Markov trace of type D which is some specialized Markov trace given in [4] such that it satisfies the Markov property. In Section 4, we define the Markov trace on $\mathcal{H}$ whose weights are determined by applying the Fourier transform to the Molien series given in Section 2. Then we prove that this trace function satisfies the Markov property.

2. The Molien series on symmetric algebra tensor exterior algebra

2.1. Let V be an n -dimensional vector space over $\mathbb{C}$ and W a finite reflection group acting on V . We denote by $S = S(V)$ the symmetric algebra of V and by $L = L(V)$ the exterior algebra of V . Let S_i (respectively L_i) be the subspace of S (respectively L) of homogeneous degree i . The actions of W on S_i and L_i are induced from the action of W on V . Let q and r be indeterminates. We define a trace function $\text{Tr}_{S \otimes L} : W \rightarrow \mathbb{C}[r][q]$ by

$$\text{Tr}_{S \otimes L}(w) = \sum_{i \geq 0} \sum_{j=0}^n \text{Tr}_{S_i \otimes L_j}(w) r^j q^i.$$

Definition 2.1. Let $\text{Irr } W$ be the set of irreducible characters of W over $\mathbb{C}$. Then we can write

$$\text{Tr}_{S \otimes L} = \sum_{\chi \in \text{Irr } W} P_{\chi}(q, r) \chi \quad \text{with } P_{\chi}(q, r) \in \mathbb{C}[r][q].$$

We call $P_{\chi}(q, r)$ the Molien series of $S \otimes L$ for χ .

Remark 2.2. Let S^W be the W -invariant subspace of S . There exists $\{f_1, f_2, \dots, f_n\}$, a set of homogeneous algebraically independent generators of S^W . Put $d_i = \deg f_i$ ($1 \leq i \leq n$). Then the set $\{d_1, d_2, \dots, d_n\}$ is independent of the choice of $\{f_1, f_2, \dots, f_n\}$ and is called the degrees of W . Let $P_W = \sum_{w \in W} q^{l(w)}$ be the Poincaré polynomial of W . Then we have a factorization

$$P_W = \prod_{i=1}^n \frac{1 - q^{d_i}}{1 - q}.$$

Let $\mathcal{I}$ be the ideal of S generated by $\{f_1, f_2, \dots, f_n\}$ and $S_W = S/\mathcal{I}$, the coinvariant algebra of W . Then S_W is also a graded W -module and its Molien series for $\chi \in \text{Irr } W$ is given by $(1 - q)^n P_W P_{\chi}(q, 0)$ which is called the fake degree of χ . We denote them by Fdeg_{χ} ($\chi \in \text{Irr } W$). For details, see [2, Chapter 11].

The next proposition is obvious from the definition of the Molien series of $S \otimes L$.

Proposition 2.3.

(1) For $w \in W$,

$$\mathrm{Tr}_{S \otimes L}(w) = \frac{\det_V(1 + rw)}{\det_V(1 - qw)}.$$

(2) For $\chi \in \mathrm{Irr} W$,

$$P_\chi(q, r) = |W|^{-1} \sum_{w \in W} \frac{\chi(w^{-1}) \det_V(1 + rw)}{\det_V(1 - qw)}.$$

In [16], Solomon has given a formula of the Molien series for the trivial character.

Theorem 2.4. Let χ_1 be the trivial character of W . Then the Molien series of $S \otimes L$ for χ_1 is given by

$$P_{\chi_1}(q, r) = \prod_{i=1}^n \frac{1 + rq^{d_i-1}}{1 - q^{d_i}},$$

where d_i ($1 \leq i \leq n$) are the degrees of W .

In [6], Gyoja, Nishiyama and Shimura have given the Molien series explicitly for each case. Here we review them.

2.2. Type A

Let $\mathfrak{S}_n$ be the n th symmetric group and V an n -dimensional vector space over $\mathbb{C}$ with basis $\{e_1, e_2, \dots, e_n\}$. The action of $\mathfrak{S}_n$ on V is induced from the permutation on $\{e_1, e_2, \dots, e_n\}$. The irreducible characters of $\mathfrak{S}_n$ are parameterized by the partitions α of n . We denote them by χ^α ($\alpha \vdash n$). For m with $0 \leq m \leq n$, we set $V_1 = \langle e_1, e_2, \dots, e_m \rangle$ and $V_2 = \langle e_{m+1}, e_{m+2}, \dots, e_n \rangle$. We embed $\mathfrak{S}_m \times \mathfrak{S}_{n-m}$ into $\mathfrak{S}_n$ such that $\mathfrak{S}_m$ acts on V_1 and $\mathfrak{S}_{n-m}$ acts on V_2 .

Lemma 2.5.

(1) For $(w_1, w_2) \in \mathfrak{S}_m \times \mathfrak{S}_{n-m} \subset \mathfrak{S}_n$, we have

$$\mathrm{Tr}_{S \otimes L}((w_1, w_2)) = \mathrm{Tr}_{S(V_1) \otimes L(V_1)}(w_1) \times \mathrm{Tr}_{S(V_2) \otimes L(V_2)}(w_2). \quad (2.2.1)$$

(2) For $\alpha \vdash m$ and $\beta \vdash n - m$, we have

$$P_{\chi^\alpha}(q, r) P_{\chi^\beta}(q, r) = \sum_{\gamma \vdash n} c_{\alpha\beta}^\gamma P_{\chi^\gamma}(q, r),$$

where $c_{\alpha\beta}^\gamma$ is the Littlewood–Richardson coefficient which is given by

$$c_{\alpha\beta}^\gamma = (\chi^\gamma |_{\mathfrak{S}_m \times \mathfrak{S}_{n-m}}, \chi^\alpha \times \chi^\beta)_{\mathfrak{S}_m \times \mathfrak{S}_{n-m}},$$

for $\gamma \vdash n$.

Proof. (1) It is obvious from the definition of $\text{Tr}_{S \otimes L}$.

(2) As we compare the two sides of (2.2.1), we have

$$\begin{aligned} \text{L.H.S. of (2.2.1)} &= \sum_{\gamma \vdash n} P_{\chi^\gamma}(q, r) \chi^\gamma((w_1, w_2)) \\ &= \sum_{\alpha \vdash m} \sum_{\beta \vdash n-m} \sum_{\gamma \vdash n} c_{\alpha\beta}^\gamma P_{\chi^\gamma}(q, r) \chi^\alpha(w_1) \chi^\beta(w_2), \end{aligned}$$

and

$$\text{R.H.S. of (2.2.1)} = \sum_{\alpha \vdash m} \sum_{\beta \vdash n-m} P_{\chi^\alpha}(q, r) P_{\chi^\beta}(q, r) \chi^\alpha(w_1) \chi^\beta(w_2).$$

By the linear independence of the irreducible characters, we obtain the result. $\square$

Here we prepare some notation. For a partition $\alpha = (\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_l) \vdash n$, we put $n(\alpha) = \sum_{i=1}^l (i-1)\alpha_i$. We identify α with its Young diagram $\{(i, j) \in \mathbb{N} \times \mathbb{N} \mid 1 \leq i \leq l, 1 \leq j \leq \alpha_i\}$. For $x = (i, j) \in \alpha$, we define the hook length by $h(x) = \alpha_i + \alpha'_j - i - j + 1$ (where α' is the dual partition of α) and the content by $c(x) = j - i$.

Theorem 2.6. For $\alpha \vdash n$, the Molien series $P_{\chi^\alpha}(q, r)$ is given by

$$P_{\chi^\alpha}(q, r) = q^{n(\alpha)} \prod_{x \in \alpha} \frac{1 + r q^{c(x)}}{1 - q^{h(x)}}.$$

Proof. It is an easy consequence of Lemma 2.5(2) and [14, I.3.Ex.3]. $\square$

Remark 2.7. For some large m such that $\alpha = (0 \leq \alpha_1 \leq \alpha_2 \leq \cdots \leq \alpha_m)$, we set $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_m)$ by $\lambda_i = \alpha_i + i - 1$ ($1 \leq i \leq m$). Then we have

$$P_{\chi^\alpha}(q, r) = \frac{(-1)^n \prod_{1 \leq i' < i \leq m} (q^{\lambda_i} - q^{\lambda_{i'}}) \prod_{x \in \alpha} (1 + r q^{c(x)})}{q^{\binom{m-1}{2} + \binom{m-2}{2} + \cdots + 1} \prod_{i=1}^m \prod_{k=1}^{\lambda_i} (q^k - 1)}$$

(cf. [14, I.1.Ex.1]).

2.3. Type B

Let W_n be the Weyl group of type B_n which is isomorphic to the semidirect product $(\mathbb{Z}/2\mathbb{Z})^n \rtimes \mathfrak{S}_n$. W_n acts on the n -dimensional vector space V such that $(\mathbb{Z}/2\mathbb{Z})^n$ acts on V by sign changes on the basis $\{e_1, e_2, \dots, e_n\}$. We recall the irreducible characters of W_n . For $\alpha \vdash n$, we define the character $\chi^{(\alpha, \emptyset)}$ by the composition of the projection $W_n \rightarrow \mathfrak{S}_n$ and the character χ^α of $\mathfrak{S}_n$. There exists a unique non-trivial character γ of $(\mathbb{Z}/2\mathbb{Z})^n$ which is invariant under the action of $\mathfrak{S}_n$. The linear character $\chi^{(\emptyset, (n))}$ is obtained by extending γ to W_n such that $\chi^{(\emptyset, (n))}(w) = 1$ for $w \in \mathfrak{S}_n$. The character $\chi^{(\emptyset, \alpha)}$ is defined by the product $\chi^{(\emptyset, (n))} \cdot \chi^{(\alpha, \emptyset)}$. For the pair of partitions (α, β) with $|\alpha| + |\beta| = n$, the character $\chi^{(\alpha, \beta)}$ is defined by inducing the character $\chi^{(\alpha, \emptyset)} \times \chi^{(\emptyset, \beta)}$ of $W_{|\alpha|} \times W_{|\beta|}$ to W_n . Now we have the set of irreducible characters

$$\text{Irr } W_n = \{ \chi^{(\alpha, \beta)} \mid (\alpha, \beta) \in \Lambda_n \},$$

where $\Lambda_n = \{(\alpha, \beta) \mid \alpha \vdash m, \beta \vdash n - m \text{ for some } m = 0, 1, \dots, n\}$.

Theorem 2.8. For $\chi^{(\alpha, \beta)} \in \text{Irr } W_n$, the Molien series $P_{\chi^{(\alpha, \beta)}}(q, r)$ is given by

$$P_{\chi^{(\alpha, \beta)}}(q, r) = q^{2n(\alpha) + 2n(\beta) + |\beta|} \prod_{x \in \alpha} \frac{1 + rq^{2c(x)+1}}{1 - q^{2h(x)}} \prod_{x \in \beta} \frac{1 + rq^{2c(x)-1}}{1 - q^{2h(x)}}.$$

Proof. By Frobenius reciprocity,

$$\begin{aligned} P_{\chi^{(\alpha, \beta)}}(q, r) &= (\text{Tr}_{S \otimes L}, \chi^{(\alpha, \beta)})_{W_n} \\ &= (\text{Tr}_{S \otimes L|_{W_{|\alpha|} \times W_{|\beta|}}, \chi^{(\alpha, \emptyset)} \times \chi^{(\emptyset, \beta)})_{W_{|\alpha|} \times W_{|\beta|}} \\ &= (\text{Tr}_{S(V_1) \otimes L(V_1)}, \chi^{(\alpha, \emptyset)})_{W_{|\alpha|}} \times (\text{Tr}_{S(V_2) \otimes L(V_2)}, \chi^{(\emptyset, \beta)})_{W_{|\beta|}}, \end{aligned}$$

where $V_1 = \langle e_1, e_2, \dots, e_{|\alpha|} \rangle$ and $V_2 = \langle e_{|\alpha|+1}, e_{|\alpha|+2}, \dots, e_n \rangle$. So the proof can be reduced to the case when either $\beta = \emptyset$ or $\alpha = \emptyset$. If $\beta = \emptyset$, then we have

$$\begin{aligned} (\text{Tr}_{S \otimes L}, \chi^{(\alpha, \emptyset)})_{W_n} &= |W_n|^{-1} \sum_{\sigma \in (\mathbb{Z}/2\mathbb{Z})^n} \sum_{w \in \mathfrak{S}_n} \chi^\alpha(w) \frac{\det(1 + rw\sigma)}{\det(1 - qw\sigma)} \\ &= |\mathfrak{S}_n|^{-1} \sum_{w \in \mathfrak{S}_n} \chi^\alpha(w) 2^{-n} \sum_{\sigma \in (\mathbb{Z}/2\mathbb{Z})^n} \frac{\det(1 + rw\sigma)}{\det(1 - qw\sigma)} \\ &= |\mathfrak{S}_n|^{-1} \sum_{w \in \mathfrak{S}_n} \chi^\alpha(w) \frac{\det(1 + qrw)}{\det(1 - q^2w)} \\ &= q^{2n(\alpha)} \prod_{x \in \alpha} \frac{1 + rq^{2c(x)+1}}{1 - q^{2h(x)}}. \end{aligned}$$

If $\alpha = \emptyset$, then we have

$$\begin{aligned}
 (\mathrm{Tr}_{S \otimes L}, \chi^{(\emptyset, \beta)})_{W_n} &= |W_n|^{-1} \sum_{\sigma \in (\mathbb{Z}/2\mathbb{Z})^n} \sum_{w \in \mathfrak{S}_n} \chi^\beta(w) \gamma(\sigma) \frac{\det(1 + rw\sigma)}{\det(1 - qw\sigma)} \\
 &= |\mathfrak{S}_n|^{-1} \sum_{w \in \mathfrak{S}_n} \chi^\beta(w) 2^{-n} \sum_{\sigma \in (\mathbb{Z}/2\mathbb{Z})^n} \frac{\gamma(\sigma) \det(1 + rw\sigma)}{\det(1 - qw\sigma)} \\
 &= |\mathfrak{S}_n|^{-1} q^n \sum_{w \in \mathfrak{S}_n} \chi^\beta(w) \frac{\det(1 + q^{-1}rw)}{\det(1 - q^2w)} \\
 &= q^{2n(\beta) + n} \prod_{x \in \beta} \frac{1 + rq^{2c(x)-1}}{1 - q^{2h(x)}}.
 \end{aligned}$$

The third equalities of both of the above are reduced to the case where w is a cyclic permutation of $e_1, e_2, \dots, e_n$. In that case, a direct computation shows that

$$\frac{\det(1 + rw\sigma)}{\det(1 - qw\sigma)} = \frac{1 - \gamma(\sigma)(-r)^n}{1 - \gamma(\sigma)q^n}$$

(cf. [10, (2.4.2)]). Hence we obtain the result. $\square$

Remark 2.9. Let $\mathrm{Fdeg}_{\chi^{(\alpha, \beta)}}$ be the fake degree of $\chi^{(\alpha, \beta)}$ and P_{W_n} the Poincaré polynomial of W_n . Then it is obvious from Remark 2.2 that

$$P_{\chi^{(\alpha, \beta)}}(q, r) = \frac{\mathrm{Fdeg}_{\chi^{(\alpha, \beta)}}}{(1 - q)^n P_{W_n}} \prod_{x \in \alpha} (1 + rq^{2c(x)+1}) \prod_{x \in \beta} (1 + rq^{2c(x)-1}).$$

2.4. Type D

Let $\tilde{W}_n$ be the Coxeter group of type D_n which is obtained as the kernel of $\chi^{(\emptyset, (n))}$, the linear character of the Coxeter group of type B_n . For $(\alpha, \beta) \in \Lambda_n$ with $\alpha \neq \beta$, the restriction of $\chi^{(\alpha, \beta)}$ to $\tilde{W}_n$ is irreducible and it is known that

$$\chi^{(\alpha, \beta)}|_{\tilde{W}_n} = \chi^{(\beta, \alpha)}|_{\tilde{W}_n}.$$

We denote it by $\chi^{\{\alpha, \beta\}}$. For $(\alpha, \alpha) \in \Lambda_n$, the restriction of $\chi^{(\alpha, \alpha)}$ to $\tilde{W}_n$ splits into two distinct irreducible characters which are denoted by $\chi^{\{\alpha, +\}}$ and $\chi^{\{\alpha, -\}}$. In this paper we do not need to distinguish them. Each irreducible character of $\tilde{W}_n$ is obtained without repetition in this way. Now we give the Molien series of $S \otimes L$ for each irreducible character of $\tilde{W}_n$.

Theorem 2.10.

(1) For $\chi^{\{\alpha, \beta\}} \in \mathrm{Irr} \tilde{W}_n$, we have

$$\begin{aligned}
P_{\chi^{\{\alpha, \beta\}}}(q, r) &= \frac{q^{2n(\alpha)+2n(\beta)}}{\prod_{x \in \alpha} (1 - q^{2h(x)}) \prod_{x \in \beta} (1 - q^{2h(x)})} \\
&\quad \times \left(q^{|\beta|} \prod_{x \in \alpha} (1 + rq^{2c(x)+1}) \prod_{x \in \beta} (1 + rq^{2c(x)-1}) \right. \\
&\quad \left. + q^{|\alpha|} \prod_{x \in \alpha} (1 + rq^{2c(x)-1}) \prod_{x \in \beta} (1 + rq^{2c(x)+1}) \right).
\end{aligned}$$

(2) For $\chi^{\{\alpha, \pm\}} \in \text{Irr } \tilde{W}_n$, we have

$$P_{\chi^{\{\alpha, \pm\}}}(q, r) = q^{4n(\alpha)+|\alpha|} \prod_{x \in \alpha} \frac{(1 + rq^{2c(x)+1})(1 + rq^{2c(x)-1})}{(1 - q^{2h(x)})^2}.$$

Proof. Let $P_{\chi^{(\alpha, \beta)}}(q, r)$ be the Molien series of type B_n . By Frobenius reciprocity, we have

$$\begin{aligned}
P_{\chi^{\{\alpha, \beta\}}}(q, r) &= P_{\chi^{(\alpha, \beta)}}(q, r) + P_{\chi^{(\beta, \alpha)}}(q, r), \\
P_{\chi^{\{\alpha, \pm\}}}(q, r) &= P_{\chi^{(\alpha, \alpha)}}(q, r).
\end{aligned}$$

We complete the proof immediately from Theorem 2.8. $\square$

Remark 2.11. Let $\text{Fdeg}_{\chi^{\{\alpha, \beta\}}}$ (respectively $\text{Fdeg}_{\chi^{\{\alpha, \pm\}}}$) be the fake degree of $\chi^{\{\alpha, \beta\}}$ (respectively $\chi^{\{\alpha, \pm\}}$) and $P_{\tilde{W}_n}$ the Poincaré polynomial of $\tilde{W}_n$. It is easily seen that

$$\begin{aligned}
P_{\tilde{W}_n} &= (1 + q^n)^{-1} P_{W_n}, \\
\text{Fdeg}_{\chi^{\{\alpha, \beta\}}} &= (1 + q^n)^{-1} q^{-|\beta|} (q^{|\alpha|} + q^{|\beta|}) \text{Fdeg}_{\chi^{(\alpha, \beta)}} \\
&= (1 + q^n)^{-1} q^{-|\alpha|} (q^{|\alpha|} + q^{|\beta|}) \text{Fdeg}_{\chi^{(\beta, \alpha)}}, \\
\text{Fdeg}_{\chi^{\{\alpha, \pm\}}} &= (1 + q^n)^{-1} \text{Fdeg}_{\chi^{(\alpha, \alpha)}}.
\end{aligned}$$

By Remark 2.9, we have

$$\begin{aligned}
P_{\chi^{\{\alpha, \beta\}}}(q, r) &= \frac{\text{Fdeg}_{\chi^{\{\alpha, \beta\}}}}{(1 - q)^n P_{\tilde{W}_n}} \frac{1}{q^{|\alpha|} + q^{|\beta|}} \left(q^{|\beta|} \prod_{x \in \alpha} (1 + rq^{2c(x)+1}) \prod_{x \in \beta} (1 + rq^{2c(x)-1}) \right. \\
&\quad \left. + q^{|\alpha|} \prod_{x \in \alpha} (1 + rq^{2c(x)-1}) \prod_{x \in \beta} (1 + rq^{2c(x)+1}) \right), \\
P_{\chi^{\{\alpha, \pm\}}}(q, r) &= \frac{\text{Fdeg}_{\chi^{\{\alpha, \pm\}}}}{(1 - q)^n P_{\tilde{W}_n}} \prod_{x \in \alpha} (1 + rq^{2c(x)+1})(1 + rq^{2c(x)-1}).
\end{aligned}$$

2.5. Exceptional type

Let (W, S) be a finite Coxeter system of exceptional type. From the data of the character table of W and the eigenvalues of each element of W for the reflection representation (for example, they are available in the CHEVIE system [3]), we can obtain the Molien series in each case except the case of type $I_2(m)$. Here we just mention the case of type $I_2(m)$ without proof. Let W be the Coxeter group of type $I_2(m)$. Using the notation in the CHEVIE system, the irreducible characters of W are as follows:

$$\text{Irr } W = \begin{cases} \{\phi_{1,0}, \phi_{1,m}, \phi_{2,1}, \phi_{2,2}, \dots, \phi_{2, \frac{m-1}{2}}\}, & \text{if } m \text{ is odd,} \\ \{\phi_{1,0}, \phi_{1,m}, \phi'_{1, \frac{m}{2}}, \phi''_{1, \frac{m}{2}}, \phi_{2,1}, \phi_{2,2}, \dots, \phi_{2, \frac{m}{2}-1}\}, & \text{if } m \text{ is even.} \end{cases}$$

Theorem 2.12. *Molien series of the irreducible characters of W are*

$$\begin{aligned} P_{\phi_{1,0}}(q, r) &= \frac{(1 + rq)(1 + rq^{m-1})}{(1 - q^2)(1 - q^m)}, \\ P_{\phi_{1,m}}(q, r) &= \frac{q^m(1 + rq^{-1})(1 + rq^{1-m})}{(1 - q^2)(1 - q^m)}, \\ P_{\phi_{2,k}}(q, r) &= \frac{(q^k + q^{m-k})(1 + rq)(1 + rq^{-1})}{(1 - q^2)(1 - q^m)}, \quad \text{for } k = 1, 2, \dots, \left[\frac{m-1}{2}\right], \\ P_{\phi'_{1, \frac{m}{2}}}(q, r) &= P_{\phi''_{1, \frac{m}{2}}}(q, r) = \frac{q^{\frac{m}{2}}(1 + rq)(1 + rq^{-1})}{(1 - q^2)(1 - q^m)} \quad (\text{if } m \text{ is even}). \end{aligned}$$

Remark 2.13.

(1) In the case of type $I_2(3) = A_2$, irreducible characters are

$$\phi_{1,0} = \chi^{(3)}, \quad \phi_{1,3} = \chi^{(1^3)} \quad \text{and} \quad \phi_{2,1} = \chi^{(2,1)}.$$

In Section 2.2, we determine the Molien series of type A_2 as W acts on $V = \langle e_1, e_2, e_3 \rangle$. Since we have a decomposition $V = V_1 \oplus V_2$, where $V_1 = \langle e_1 + e_2 + e_3 \rangle$ affords the trivial representation and $V_2 = \langle e_1 - e_2, e_2 - e_3 \rangle$ affords the reflection representation. Hence if you multiply the Molien series in Theorem 2.12 by $(1 + r)/(1 - q)$, then you obtain the Molien series in Theorem 2.6.

(2) In the case of type $I_2(4) = B_2$, irreducible characters are

$$\begin{aligned} \phi_{1,0} &= \chi^{((2), \cdot)}, & \phi_{1,4} &= \chi^{(\cdot, (1^2))}, & \phi'_{1,2} &= \chi^{((1^2), \cdot)}, \\ \phi''_{1,2} &= \chi^{(\cdot, (2))} & \text{and} & & \phi_{2,1} &= \chi^{((1), (1))}. \end{aligned}$$

You can see that Theorem 2.12 gives the same formula as in Theorem 2.8.

3. The Markov trace

3.1. Type A

Let $\mathcal{H}(A_{n-1})$ be the Iwahori–Hecke algebra of type A_{n-1} over the field of rational functions $\mathbb{Q}(q)$. The algebra $\mathcal{H}(A_{n-1})$ has a presentation with generators $T_1, T_2, \dots, T_{n-1}$ and the defining relations:

$$\begin{aligned} T_i^2 &= q + (q-1)T_i, & \text{for } 1 \leq i \leq n-1, \\ T_i T_{i+1} T_i &= T_{i+1} T_i T_{i+1}, & \text{for } 1 \leq i \leq n-2, \\ T_i T_j &= T_j T_i, & \text{if } |i-j| \geq 2, \text{ for } 1 \leq i, j \leq n-1. \end{aligned}$$

We have natural embeddings $\mathcal{H}(A_1) \subset \mathcal{H}(A_2) \subset \dots$. Let z be another indeterminate. Then we have a trace function on $\mathcal{H}(A_{n-1})$ called the Markov trace which is constructed in [8].

Theorem 3.1. *There exists a unique $\mathbb{Q}(q)$ -linear function $\tau_{q;z}^{A_{n-1}} : \mathcal{H}(A_{n-1}) \rightarrow \mathbb{Q}(q, z)$, which satisfies:*

- (1) $\tau_{q;z}^{A_{n-1}}(1) = 1$,
- (2) $\tau_{q;z}^{A_{n-1}}|_{\mathcal{H}(A_{n-2})} = \tau_{q;z}^{A_{n-2}}$,
- (3) $\tau_{q;z}^{A_{n-1}}(ab) = \tau_{q;z}^{A_{n-1}}(ba)$, for any $a, b \in \mathcal{H}(A_{n-1})$,
- (4) $\tau_{q;z}^{A_{n-1}}(aT_{n-1}) = z\tau_{q;z}^{A_{n-1}}(a)$, for any $a \in \mathcal{H}(A_{n-2})$.

The algebra $\mathcal{H}(A_{n-1})$ is split semisimple and its irreducible characters are in canonical bijection with those of $\mathfrak{S}_n$ via specialization. We denote by χ_q^α the irreducible character of $\mathcal{H}(A_{n-1})$ corresponding to the character χ^α of $\mathfrak{S}_n$. Then we can write

$$\tau_{q;z}^{A_{n-1}} = \sum_{\alpha \vdash n} w_{q;z}^\alpha \chi_q^\alpha \quad \text{with } w_{q;z}^\alpha \in \mathbb{Q}(q, z).$$

The $w_{q;z}^\alpha$ are called weights of $\tau_{q;z}^{A_{n-1}}$. Ocneanu has given the formula of the weight $w_{q;z}^\alpha$ which is described in [8].

Theorem 3.2.

$$w_{q;z}^\alpha = q^{n(\alpha)} \prod_{x \in \alpha} \frac{1 - q + (1 - q^{c(x)})z}{1 - q^{h(x)}}.$$

3.2. Type B

Let $\mathcal{H}(B_n)$ be the Iwahori–Hecke algebra of type B_n over the field $\mathbb{Q}(Q, q)$. The algebra $\mathcal{H}(B_n)$ has a presentation with generators $T_0, T_1, T_2, \dots, T_{n-1}$ and the defining relations:

$$\begin{aligned}
T_0^2 &= Q + (Q - 1)T_0, \\
T_i^2 &= q + (q - 1)T_i, \quad \text{for } 1 \leq i \leq n - 1, \\
T_0 T_1 T_0 T_1 &= T_1 T_0 T_1 T_0, \\
T_i T_{i+1} T_i &= T_{i+1} T_i T_{i+1}, \quad \text{for } 1 \leq i \leq n - 2, \\
T_i T_j &= T_j T_i, \quad \text{if } |i - j| \geq 2, \quad \text{for } 0 \leq i, j \leq n - 1.
\end{aligned}$$

We have natural embeddings $\mathcal{H}(B_1) \subset \mathcal{H}(B_2) \subset \cdots$. For two indeterminates y and z , Geck and Lambropoulou have constructed a trace function on $\mathcal{H}(B_n)$ called the Markov trace of type B_n in [4].

Theorem 3.3. *There exists a unique $\mathbb{Q}(Q, q)$ -linear function $\tau_{Q,q;y,z}^{B_n} : \mathcal{H}(B_n) \rightarrow \mathbb{Q}(Q, q, y, z)$, which satisfies:*

- (1) $\tau_{Q,q;y,z}^{B_n}(1) = 1$,
- (2) $\tau_{Q,q;y,z}^{B_n}|_{\mathcal{H}(B_{n-1})} = \tau_{Q,q;y,z}^{B_{n-1}}$,
- (3) $\tau_{Q,q;y,z}^{B_n}(ab) = \tau_{Q,q;y,z}^{B_n}(ba)$, for any $a, b \in \mathcal{H}(B_n)$,
- (4) $\tau_{Q,q;y,z}^{B_n}(aT_{n-1}) = z\tau_{Q,q;y,z}^{B_n}(a)$, for any $a \in \mathcal{H}(B_{n-1})$ if $n \geq 2$,
- (5) $\tau_{Q,q;y,z}^{B_n}(T'_0 T'_1 \cdots T'_{n-1}) = y^n$,

where $T'_i = T_i T_{i-1} \cdots T_1 T_0 T_1^{-1} \cdots T_{i-1}^{-1} T_i^{-1}$ ($0 \leq i \leq n - 1$).

Remark 3.4. When we consider $\mathcal{H}(A_{n-1})$ as a subalgebra of $\mathcal{H}(B_n)$, the restriction of $\tau_{Q,q;y,z}^{B_n}$ to $\mathcal{H}(A_{n-1})$ coincides with $\tau_{q;z}^{A_{n-1}}$.

The algebra $\mathcal{H}(B_n)$ is split semisimple and its irreducible characters are in canonical bijection with those of W_n , the Coxeter group of type B_n . We denote by $\chi_{Q,q}^{(\alpha,\beta)}$ the irreducible character of $\mathcal{H}(B_n)$ corresponding to $\chi^{(\alpha,\beta)}$. Then we can write

$$\tau_{Q,q;y,z}^{B_n} = \sum_{(\alpha,\beta) \in \Lambda_n} w_{Q,q;y,z}^{(\alpha,\beta)} \chi_{Q,q}^{(\alpha,\beta)} \quad \text{with } w_{Q,q;y,z}^{(\alpha,\beta)} \in \mathbb{Q}(Q, q, y, z).$$

For $(\alpha, \beta) \in \Lambda_n$, we choose some large m such that $\alpha = (0 \leq \alpha_1 \leq \alpha_2 \leq \cdots \leq \alpha_{m+1})$ and $\beta = (0 \leq \beta_1 \leq \beta_2 \leq \cdots \leq \beta_m)$ by adding some zeros as parts of α or β , if necessary. Then we define a symbol of $\chi^{(\alpha,\beta)}$ by

$$\begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_{m+1} \\ \mu_1 & \mu_2 & \cdots & \mu_m \end{pmatrix},$$

where $\lambda_i = \alpha_i + i - 1$ ($1 \leq i \leq m+1$) and $\mu_j = \beta_j + j - 1$ ($1 \leq j \leq m$). Finally we define

$$D_{Q,q}^{(\alpha,\beta)} = \frac{\prod_{1 \leq i' < i \leq m+1} (q^{\lambda_i} - q^{\lambda_{i'}}) \prod_{1 \leq j' < j \leq m} (q^{\mu_j} - q^{\mu_{j'}}) \prod_{i=1}^{m+1} \prod_{j=1}^m (q^{\lambda_i-1} Q + q^{\mu_j})}{q^{f(m)} (Qq^{-1})^{\binom{m}{2}} (1+Qq^{-1})^m \prod_{i=1}^{m+1} \prod_{h=1}^{\lambda_i} (q^h - 1) (q^{h-1} Q + 1) \prod_{j=1}^m \prod_{h=1}^{\mu_j} (q^h - 1) (q^{h+1} Q^{-1} + 1)},$$

where

$$f(m) = \binom{2m-1}{2} + \binom{2m-3}{2} + \cdots + \binom{3}{2} = \frac{m(m-1)(4m+1)}{6}.$$

We note that $D_{Q,q}^{(\alpha,\beta)}$ is well defined, that is, the right-hand side of the equation does not depend on the choice of m (cf. [10]). Iancu has given the formula of the weight $w_{Q,q;y,z}^{(\alpha,\beta)}$ in [7].

Theorem 3.5. For $(\alpha, \beta) \in \Lambda_n$, we define

$$\begin{aligned} R_{Q,q;y,z}^{(\alpha,\beta)} &= \prod_{x \in \alpha} ((q-1)(1+q^{c(x)}y) + (q^{c(x)}-1)(1+Qq^{c(x)}z)) \\ &\quad \times \prod_{x \in \beta} ((q-1)(1-Q^{-1}q^{c(x)}y) + (q^{c(x)}-1)(1+Q^{-1}q^{c(x)}z)). \end{aligned}$$

Then we have

$$w_{Q,q;y,z}^{(\alpha,\beta)} = D_{Q,q}^{(\alpha,\beta)} R_{Q,q;y,z}^{(\alpha,\beta)}.$$

Remark 3.6.

(1) If we specialize $Q \mapsto q$ and $y \mapsto z$, then we have

$$D_{q,q}^{(\alpha,\beta)} = \frac{\prod_{1 \leq i' < i \leq m+1} (q^{\lambda_i} - q^{\lambda_{i'}}) \prod_{1 \leq j' < j \leq m} (q^{\mu_j} - q^{\mu_{j'}}) \prod_{i=1}^{m+1} \prod_{j=1}^m (q^{\lambda_i} + q^{\mu_j})}{2^m q^{f(m)} \prod_{i=1}^{m+1} \prod_{h=1}^{\lambda_i} (q^{2h} - 1) \prod_{j=1}^m \prod_{h=1}^{\mu_j} (q^{2h} - 1)}$$

and

$$R_{q,q;z,z}^{(\alpha,\beta)} = \prod_{x \in \alpha} ((q-1) + (q^{2c(x)+1} - 1)z) \prod_{x \in \beta} ((q-1) + (q^{2c(x)-1} - 1)z).$$

Let $P_W = \frac{\prod_{i=1}^n (q^{2i} - 1)}{(q-1)^n}$ be the Poincaré polynomial with respect to the Coxeter group of type B_n . Then $(q-1)^n P_W \times D_{q,q}^{(\alpha,\beta)}$ is called the generic degree of $\chi_{q,q}^{(\alpha,\beta)}$ which we denote by $\text{Gdeg}_{\chi_{q,q}^{(\alpha,\beta)}}$ (see Remark 3.8(3)).

(2) If we specialize $Q \mapsto 1$, then we have

$$D_{1,q}^{(\alpha,\beta)} = \frac{\prod_{1 \leq i' < i \leq m} (q^{\lambda_i-1} - q^{\lambda_{i'}-1}) \prod_{1 \leq j' < j \leq m} (q^{\mu_j} - q^{\mu_{j'}}) \prod_{i=1}^m \prod_{j=1}^m (q^{\lambda_i-1} + q^{\mu_j})}{2^m q^{\binom{2m-2}{2} + \binom{2m-4}{2} + \dots} \prod_{i=1}^m \prod_{h=1}^{\lambda_i-1} (q^{2h} - 1) \prod_{j=1}^m \prod_{h=1}^{\mu_j} (q^{2h} - 1)}$$

and

$$\begin{aligned} R_{1,q;y,z}^{(\alpha,\beta)} &= \prod_{x \in \alpha} ((q-1)(1+q^{c(x)}y) + (q^{2c(x)}-1)z) \\ &\quad \times \prod_{x \in \beta} ((q-1)(1-q^{c(x)}y) + (q^{2c(x)}-1)z) \\ &= R_{1,q;-y,z}^{(\beta,\alpha)}. \end{aligned}$$

3.3. Markov property

Now we have the Markov traces of type A_n and of type B_n . It is a natural question whether there exist trace functions of other types which should be called Markov traces. First we give a definition of a Markov trace for a general Iwahori–Hecke algebra based on common properties of both Markov traces of type A_n and of type B_n . Let (W, S) be a finite Coxeter system and $\mathcal{H}_S$ the corresponding Iwahori–Hecke algebra over a field K with generators $\{T_s\}_{s \in S}$ and parameters $\{q_s\}_{s \in S} \subset K$. For $I \subset S$, we have a canonical embedding of the Iwahori–Hecke algebras $\mathcal{H}_I \subset \mathcal{H}_S$ [5, 4.4.7].

Definition 3.7. Let $\tau: \mathcal{H}_S \rightarrow K$ be a K -linear function and $z_s \in K$ for $s \in S$. Then τ is called a Markov trace (with parameters $\{z_s\}_{s \in S}$) if the following conditions are satisfied.

- (M1) $\tau(1) = 1$.
- (M2) $\tau(ab) = \tau(ba)$, for $a, b \in \mathcal{H}_S$.
- (M3) $\tau(aT_s) = z_s \tau(a)$, for $s \in S$ and for $a \in \mathcal{H}_{S \setminus \{s\}}$.

We call the conditions above the Markov property.

Remark 3.8.

- (1) If there exists a trace function which satisfies the Markov property, then $z_s = z_{s'}$ if s and s' are conjugate in W .
- (2) A K -linear function $\tau: \mathcal{H}_S \rightarrow K$ is uniquely determined by the Markov property only when the Coxeter system satisfies

$$W = \bigcup_{s \in S} (W_{S \setminus \{s\}} \cup W_{S \setminus \{s\}} s W_{S \setminus \{s\}}).$$

It is easily seen that the type A_n ($n \geq 1$) are the only cases which satisfy this condition.

- (3) Let τ_0 be the canonical symmetrizing trace of $\mathcal{H}_S$ which is defined by $\tau_0(T_w) = \delta_{w,1}$ for $w \in W$. Then τ_0 satisfies the Markov property with $z_s = 0$ for all $s \in S$. Take a base field over which $\mathcal{H}_S$ is split semisimple and let $\text{Irr } \mathcal{H}_S$ be the set of irreducible characters of $\mathcal{H}_S$. Then we can write

$$\tau_0 = \sum_{\chi \in \text{Irr } \mathcal{H}_S} w_0^\chi \chi.$$

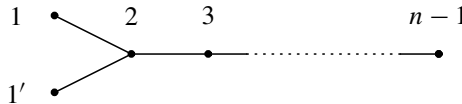
Let P_W be the Poincaré polynomial of W . Then $P_W w_0^\chi$ coincides with Gdeg_χ , the generic degree of χ (cf. [5, 8.1.8]).

Theorem 3.9. *Let (W, S) be the Coxeter system of type A_{n-1} or B_n . Then the Markov trace defined above satisfies the Markov property with $z_{s_1} = z_{s_2} = \cdots = z_{s_{n-1}} = z$ and $z_{s_0} = y$.*

In case of type A_{n-1} , it can be proved directly. In case of type B_n , Geck and Lambropoulou have given a proof in [4, §4].

3.4. Type D

In [4], Geck and Lambropoulou have constructed the Markov trace of type D_n . But it does not satisfy the Markov property in the sense of Definition 3.7. In order to satisfy the Markov property, it should be specialized appropriately. We construct $\mathcal{H}(D_n)$, the Iwahori–Hecke algebra of type D_n as a subalgebra of $\mathcal{H}(B_n)_{Q=1}$, the Iwahori–Hecke algebra of type B_n specialized by $Q \mapsto 1$. Let $\{T_0, T_1, \dots, T_{n-1}\}$ be the generators of $\mathcal{H}(B_n)_{Q=1}$ given as above and put $T_{1'} = T_0 T_1 T_0$. Then $\{T_{1'}, T_1, T_2, \dots, T_{n-1}\}$ generate a subalgebra which is isomorphic to $\mathcal{H}(D_n)$. The corresponding diagram is as follows.



Let $\tau_{1,q;y,z}^{B_n}$ be the Markov trace of $\mathcal{H}(B_n)_{Q=1}$. Here we consider the restriction of $\tau_{1,q;y,z}^{B_n}$ to $\mathcal{H}(D_n)$. It is obvious that $\tau_{1,q;y,z}^{B_n}|_{\mathcal{H}(D_n)}$ satisfies the conditions (M1) and (M2). If $s = s_2, s_3, \dots, s_{n-1}$, then the equations of (M3) are satisfied from the Markov property of $\tau_{Q,q;y,z}^{B_n}$. When $s = s_1$ or $s_{1'}$, the equations of (M3) are reduced to the equation $\tau_{1,q;y,z}^{B_n}(T_{1'} T_1) = z^2$. Since we have

$$\begin{aligned} \tau_{1,q;y,z}^{B_n}(T_{1'} T_1) &= \tau_{1,q;y,z}^{B_n}(T_0 T_1 T_0 T_1) = q \tau_{1,q;y,z}^{B_n}(T_0 T_1 T_0 T_1^{-1}) + (q-1) \tau_{1,q;y,z}^{B_n}(T_1) \\ &= qy^2 + (q-1)z, \end{aligned}$$

it is required that $y^2 = q^{-1}z^2 + (q^{-1} - 1)z$. It is easily seen that $\tau_{1,q;y,z}^{B_n}(T_w) \in \mathbb{Q}[q, z, y^2]$ for any element $w \in W(D_n)$. So we define $\tau_{q;z}^{D_n}$, the Markov trace of $\mathcal{H}(D_n)$ by the specialization of $\tau_{1,q;y,z}^{B_n}|_{\mathcal{H}(D_n)}$ through $y^2 \mapsto q^{-1}z^2 + (q^{-1} - 1)z$. Let $\chi_q^{\{\alpha,\beta\}}$ (respectively

$\chi_q^{\{\alpha, \pm\}}$ be the irreducible character of $\mathcal{H}(D_n)$ corresponding to the irreducible character $\chi^{\{\alpha, \beta\}}$ (respectively $\chi^{\{\alpha, \pm\}}$) of $\tilde{W}_n$. Then we can write

$$\tau_{q;z}^{D_n} = \sum w_{q;z}^{\{\alpha, \beta\}} \chi_q^{\{\alpha, \beta\}} + \sum w_{q;z}^{\{\alpha, +\}} \chi_q^{\{\alpha, +\}} + \sum w_{q;z}^{\{\alpha, -\}} \chi_q^{\{\alpha, -\}},$$

with $w_{q;z}^{\{\alpha, \beta\}}, w_{q;z}^{\{\alpha, \pm\}} \in \mathbb{Q}(q, z)$.

For a pair of partitions (α, β) , we choose some large m such that $\alpha = (0 \leq \alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_m)$ and $\beta = (0 \leq \beta_1 \leq \beta_2 \leq \dots \leq \beta_m)$ and define a symbol of $\chi^{\{\alpha, \beta\}}$ (or $\chi^{\{\alpha, \pm\}}$ if $\alpha = \beta$) by

$$\begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_m \\ \mu_1 & \mu_2 & \cdots & \mu_m \end{pmatrix},$$

where $\lambda_i = \alpha_i + i - 1$ ($1 \leq i \leq m$) and $\mu_j = \beta_j + j - 1$ ($1 \leq j \leq m$). Then we define $D_q^{\{\alpha, \beta\}}$ (or $D_q^{\{\alpha, \pm\}} = D_q^{\{\alpha, \alpha\}}$ if $\alpha = \beta$) by

$$D_q^{\{\alpha, \beta\}} = \frac{\prod_{1 \leq i' < i \leq m} (q^{\lambda_i} - q^{\lambda_{i'}}) \prod_{1 \leq j' < j \leq m} (q^{\mu_j} - q^{\mu_{j'}}) \prod_{i=1}^m \prod_{j=1}^m (q^{\lambda_i} + q^{\mu_j})}{2^c q^{\binom{2m-2}{2} + \binom{2m-4}{2} + \dots} \prod_{i=1}^m \prod_{h=1}^{\lambda_i} (q^{2h} - 1) \prod_{j=1}^m \prod_{h=1}^{\mu_j} (q^{2h} - 1)},$$

where

$$c = \begin{cases} m-1, & \text{if } \alpha \neq \beta, \\ m, & \text{if } \alpha = \beta. \end{cases}$$

We note that $D_q^{\{\alpha, \beta\}}$ is well defined, that is, the right-hand side of the equation does not depend on the choice of m (cf. [10]). Let $P_{\tilde{W}_n} = \frac{(q^n - 1) \prod_{i=1}^{n-1} (q^{2i} - 1)}{(q-1)^n}$ be the Poincaré polynomial with respect to the Coxeter group of type D_n . Then $(q-1)^n P_{\tilde{W}_n} D_q^{\{\alpha, \beta\}}$ (respectively $(q-1)^n P_{\tilde{W}_n} D_q^{\{\alpha, \pm\}}$) is equal to the generic degree $\text{Gdeg}_{\chi_q^{\{\alpha, \beta\}}}$ (respectively $\text{Gdeg}_{\chi_q^{\{\alpha, \pm\}}}$) of $\chi_q^{\{\alpha, \beta\}}$ (respectively $\chi_q^{\{\alpha, \pm\}}$) (see Remark 3.8(3)). It is easily seen by Remark 3.6 that

$$D_q^{\{\alpha, \beta\}} = 2D_{1,q}^{(\alpha, \beta)} = 2D_{1,q}^{(\beta, \alpha)},$$

$$D_q^{\{\alpha, \pm\}} = D_{1,q}^{(\alpha, \alpha)}.$$

Now we give a formula of the weight of $\tau_{q;z}^{D_n}$.

Theorem 3.10. For $\chi_q^{\{\alpha, \beta\}}$ (respectively $\chi_q^{\{\alpha, \pm\}}$), we define $R_{q;z}^{\{\alpha, \beta\}}$ (respectively $R_{q;z}^{\{\alpha, \pm\}}$) by the specialization of $R_{1,q;y,z}^{(\alpha, \beta)} + R_{1,q;y,z}^{(\beta, \alpha)}$ (respectively $R_{1,q;y,z}^{(\alpha, \alpha)}$) through $y^2 \mapsto q^{-1}z^2 + (q^{-1} - 1)z$. Then we have:

$$(1) \quad w_{q;z}^{\{\alpha, \beta\}} = D_q^{\{\alpha, \beta\}} R_{q;z}^{\{\alpha, \beta\}}.$$

$$\begin{aligned}
 (2) \quad w_{q;z}^{\{\alpha,\pm\}} &= D_q^{\{\alpha,\pm\}} R_{q;z}^{\{\alpha,\pm\}} \\
 &= \frac{\text{Gdeg}_{\chi_q^{\{\alpha,\pm\}}} \tau_{q;z}^{D_n}}{(q-1)^n P_{\tilde{W}_n}} \prod_{x \in \alpha} ((q-1) + (q^{2c(x)+1} - 1)z)((q-1) + (q^{2c(x)-1} - 1)z).
 \end{aligned}$$

Proof. By the definition of $\tau_{q;z}^{D_n}$ and the induction law of the irreducible characters of $\mathcal{H}(D_n)$ and those of $\mathcal{H}(B_n)$, we have

$$\begin{aligned}
 w_{q;z}^{\{\alpha,\beta\}} &= (w_{1,q;y,z}^{(\alpha,\beta)} + w_{1,q;y,z}^{(\beta,\alpha)})|_{y^2=q^{-1}z^2+(q^{-1}-1)z}, \\
 w_{q;z}^{\{\alpha,\pm\}} &= w_{1,q;y,z}^{(\alpha,\alpha)}|_{y^2=q^{-1}z^2+(q^{-1}-1)z}.
 \end{aligned}$$

The second equality of (2) is an easy consequence of Remark 3.6. $\square$

4. Fourier transforms

4.1. We sketch the Fourier transform defined by Lusztig. For details, we refer to [2] or [12]. Let G be a connected reductive algebraic group defined over the finite field with q elements and $F: G \rightarrow G$ a Frobenius map. We assume that G^F is split. Let W be the Weyl group of G with respect to some F -stable maximally split torus. The set of unipotent characters of G^F is decomposed into the so-called families. For each family $\mathcal{F}$, let $\mathcal{M}(\mathcal{F})$ be an index set of $\mathcal{F}$ and we denote by ρ_x the unipotent character labeled by $x \in \mathcal{M}(\mathcal{F})$. Let $\mathcal{H}$ be the Hecke algebra with respect to the Borel subgroup of G^F , and $\text{Irr } W \rightarrow \text{Irr } \mathcal{H}$ ($\chi \mapsto \chi_q$) the canonical bijection between the sets of irreducible characters. Then we have a natural inclusion map $\iota: \text{Irr } W \rightarrow \bigsqcup_{\mathcal{F}} \mathcal{M}(\mathcal{F})$ such that

$$\rho_{\iota(\chi)}|_{\mathcal{H}} = \chi_q, \quad \chi \in \text{Irr } W.$$

For $\chi \in \text{Irr } W$, let R_χ be the so-called almost character (cf. [2, 12.3]) of G^F which is a linear combination of the Deligne–Lusztig characters. Then Lusztig has constructed the Fourier transform matrix $(\{x, x'\})_{(x,x') \in \mathcal{M}(\mathcal{F}) \times \mathcal{M}(\mathcal{F})}$ which satisfies

$$R_\chi = \sum_{x' \in \mathcal{M}(\mathcal{F})} \{\iota(\chi), x'\} \rho_{x'}, \quad \text{for } \chi \in \text{Irr } W \quad \text{with } \iota(\chi) \in \mathcal{M}(\mathcal{F}).$$

For $x \in \mathcal{M}(\mathcal{F})$, we define a generalized character R_x of G^F by

$$R_x = \sum_{x' \in \mathcal{M}(\mathcal{F})} \{x, x'\} \rho_{x'}.$$

It is known that

$$\deg R_x = \begin{cases} \text{Fdeg}_\chi, & \text{if } \iota(\chi) = x, \\ 0, & \text{if } x \notin \iota(\text{Irr } W), \end{cases}$$

and $\deg \rho_{\iota(\chi)} = \text{Gdeg}_{\chi_q}$ for $\chi \in \text{Irr } W$. Since the Fourier transform matrix is involutive, we obtain the following formula:

$$\text{Gdeg}_{\chi_q} = \sum_{\chi' \in \text{Irr } W} \{\iota(\chi), \iota(\chi')\} \text{Fdeg}_{\chi'}, \quad \chi \in \text{Irr } W, \quad (4.1.1)$$

where $\{\iota(\chi), \iota(\chi')\} = 0$ if χ and χ' are not contained in a common family. The Fourier transforms have been also constructed for the non-crystallographic cases and they satisfy the equalities (4.1.1) (cf. [1,13,15]).

4.2. In the rest of this section, we show that the Fourier transform above sends the Molien series determined in Section 2 to the weights of the Markov trace.

Let (W, S) be a finite Coxeter system and $\mathcal{H}$ the corresponding Iwahori–Hecke algebra with parameter q . Let $P_{\chi}(q, r)$ ($\chi \in \text{Irr } W$) be the Molien series determined in Section 2 and define $w^{\chi} \in \mathbb{C}(q, z)$ by

$$w^{\chi} = \left(\frac{1-q}{1+r} \right)^n \sum_{\chi' \in \text{Irr } W} \{\iota(\chi), \iota(\chi')\} P_{\chi'}(q, r), \quad (4.2.1)$$

where $\{\iota(\chi), \iota(\chi')\}$ is the number determined from the Fourier transform.

Theorem 4.1. *Determine a trace function τ on $\mathcal{H}$ by*

$$\tau = \sum_{\chi \in \text{Irr } W} w^{\chi} \chi_q.$$

Then we have:

- (1) τ satisfies the Markov property with $z_s = (q-1)r/(1+r)$ for all $s \in S$.
- (2) Let I be a subset of S and $\mathcal{H}_I$ the corresponding parabolic subalgebra of $\mathcal{H}$. Determine a trace function τ_I on $\mathcal{H}_I$ as τ above. Then we have

$$\tau|_{\mathcal{H}_I} = \tau_I. \quad (4.2.2)$$

Remark 4.2. Put $z = (q-1)r/(1+r)$. By (4.1.1) and Remark 2.2, the weight w^{χ} of τ specializes to Gdeg_{χ}/P_W by $z \mapsto 0$ where P_W is the Poincaré polynomial of W . Hence the specialization of τ by $z \mapsto 0$ coincides with τ_0 , the canonical symmetrizing trace of $\mathcal{H}$ (cf. [5, Chapter 8]).

The theorem is reduced to the case when (W, S) is irreducible. We prove the theorem by a case-by-case argument. For exceptional cases except the cases of type $I_2(m)$, we can check the theorem by using the CHEVIE system. In fact, to prove the second statement of the theorem, you just compare the values of both sides of (4.2.2) for the elements of minimal length (cf. [5, 8.2.6]). To prove that τ satisfies the condition (M3) in Definition 3.7,

you have to check the equality for any pair $s \in S$ and $a = T_w$ ($w \in W_{S \setminus \{s\}}$). If you use the following simple tools, then you can reduce the number of cases which you should check.

- (I) Let $s \in S$, $w \in W_{S \setminus \{s\}}$ and $t_1, t_2, \dots, t_k \in S \setminus \{s\}$ which all commute with s . Put $w_i = t_i \dots t_1 w t_1 \dots t_i$ for $i = 1, 2, \dots, k$. If $l(w) = l(w_1) = \dots = l(w_{k-1}) > l(w_k)$, then we have $\tau(T_w T_s) = \tau(T_{w_1} T_s) = \dots = \tau(T_{w_{k-1}} T_s) = (q-1)\tau(T_{w_k t_k} T_s) + q\tau(T_{w_k} T_s)$. Hence $\tau(T_w T_s) = z\tau(T_w)$ is reduced to $\tau(T_{w_k} T_s) = z\tau(T_{w_k})$ and $\tau(T_{w_k t_k} T_s) = z\tau(T_{w_k t_k})$, which are shorter cases than w .
- (II) Let $s, t \in S$ and $w = w_1 t w_2$ with $w_1, w_2 \in W_{S \setminus \{s, t\}}$ and $l(w) = l(w_1) + l(w_2) + 1$. If we have $\tau(T_{w_2} T_s T_{w_1} T_t) = z\tau(T_{w_2} T_s T_{w_1})$, then $\tau(T_w T_s) = \tau(T_{w_1} T_t T_{w_2} T_s) = z\tau(T_{w_1} T_{w_2} T_s)$. Hence $\tau(T_w T_s) = z\tau(T_w)$ is reduced to $\tau(T_{w_2} T_{w_1} T_t) = z\tau(T_{w_2} T_{w_1})$ and $\tau(T_{w_1} T_{w_2} T_s) = z\tau(T_{w_1} T_{w_2})$.
- (III) Let $s, t \in S$ with $sts = tst$ and $w' \in W_{S \setminus \{s, t\}}$. Put $w = tw't$ and assume that $l(w) = l(w') + 2$ and $sw' = w's$. Then we have $\tau(T_w T_s) = \tau(T_t T_{w'} T_t T_s) = (q-1)\tau(T_{w'} T_s) + q\tau(T_{w'} T_t)$. Hence $\tau(T_w T_s) = z\tau(T_w)$ is reduced to $\tau(T_{w'} T_s) = z\tau(T_{w'})$ and $\tau(T_{w'} T_t) = z\tau(T_{w'})$.

Then you can see that the equations of (M3) are satisfied for all reduced cases of exceptional type. If you need the weights of τ or the values of τ in the cases of exceptional type, then you can obtain them from me on demand.

Now we prove the theorem in the cases of type A_{n-1} , B_n , D_n and $I_2(m)$.

4.3. Type A

In this case, every family consists of a single element and the Fourier transform matrix is trivial. So what we have to do is to check that the weight $w_{q,z}^\alpha$ with $z = (q-1)r/(1+r)$ given in Theorem 3.2 is equal to $((1-q)/(1+r))^n P_{\chi^\alpha}(q, r)$ for any $\alpha \vdash n$. Although direct calculation gives a proof, here we prove the theorem without a result of Ocneanu but using Starkey's rule.

Theorem 4.3 (Starkey's rule). *For two partitions α and β of n , we have*

$$\chi_q^\alpha(T_{w_\beta}) = |\mathfrak{S}_\beta|^{-1} (q-1)^{-\beta'_1} \sum_{w \in \mathfrak{S}_\beta} \chi^\alpha(w) \det(q-w),$$

where β' is the dual partition of β , $\mathfrak{S}_\beta = \mathfrak{S}_{\beta_1} \times \mathfrak{S}_{\beta_2} \times \dots \times \mathfrak{S}_{\beta_{\beta'_1}}$ and w_β is the Coxeter element of $\mathfrak{S}_\beta$.

By this theorem, we can compute character values of $\mathcal{H}(\mathfrak{S}_n)$ purely using the data of $\mathfrak{S}_n$. For a proof, see [5, 9.2.11].

Now we prove Theorem 4.1 in the case of type A_{n-1} . It is sufficient to check the following equality:

$$\sum_{\alpha \vdash n} \left(\frac{1-q}{1+r} \right)^n P_{\chi^\alpha}(q, r) \chi_q^\alpha(T_{w_\beta}) = \left(\frac{(q-1)r}{1+r} \right)^{n-\beta'_1}. \quad (4.3.1)$$

$$\begin{aligned}
\text{L.H.S. of (4.3.1)} &= \left(\frac{1-q}{1+r}\right)^n \sum_{\alpha \vdash n} |\mathfrak{S}_n|^{-1} \sum_{v \in \mathfrak{S}_n} \frac{\chi^\alpha(v) \det(1+rv)}{\det(1-qv)} \\
&\quad \times |\mathfrak{S}_\beta|^{-1} (q-1)^{-\beta'_1} \sum_{w \in \mathfrak{S}_\beta} \chi^\alpha(w) \det(q-w) \\
&= \left(\frac{1-q}{1+r}\right)^n (q-1)^{-\beta'_1} |\mathfrak{S}_\beta|^{-1} \sum_{w \in \mathfrak{S}_\beta} \sum_{v \in \mathfrak{S}_n} \frac{\det(1+rv) \det(q-w)}{\det(1-qv)} \\
&\quad \times |\mathfrak{S}_n|^{-1} \sum_{\alpha \vdash n} \chi^\alpha(v) \chi^\alpha(w)
\end{aligned}$$

(by the second orthogonality relations)

$$\begin{aligned}
&= \left(\frac{1-q}{1+r}\right)^n (q-1)^{-\beta'_1} |\mathfrak{S}_\beta|^{-1} \sum_{w \in \mathfrak{S}_\beta} \frac{\det(1+rw) \det(q-w)}{\det(1-qw)} \\
&= \left(\frac{q-1}{1+r}\right)^n (q-1)^{-\beta'_1} |\mathfrak{S}_\beta|^{-1} \sum_{w \in \mathfrak{S}_\beta} \det(r+w).
\end{aligned}$$

Since $\mathfrak{S}_\beta$ -module V decomposes as

$$V = V_1 \oplus V_{2,1} \oplus V_{2,2} \oplus \cdots \oplus V_{2,\beta'_1},$$

where V_1 is a β'_1 -dimensional trivial module and $V_{2,i}$ is a $(\beta_i - 1)$ -dimensional vector space which affords the reflection representation of $\mathfrak{S}_{\beta_i} \subset \mathfrak{S}_\beta$ for $1 \leq i \leq \beta'_1$, we have

$$\begin{aligned}
\text{L.H.S. of (4.3.1)} &= \left(\frac{q-1}{1+r}\right)^n (q-1)^{-\beta'_1} (1+r)^{\beta'_1} \prod_{i=1}^{\beta'_1} |\mathfrak{S}_{\beta_i}|^{-1} \sum_{w \in \mathfrak{S}_{\beta_i}} \det(r+w) \\
&= \left(\frac{(q-1)r}{1+r}\right)^{n-\beta'_1}.
\end{aligned}$$

The last equality comes from the fact that for $1 \leq i \leq \beta'_1$ and for $0 \leq j \leq \beta_i - 1$, the j th exterior power of $V_{2,i}$ are irreducible and pairwise non-isomorphic as $\mathfrak{S}_{\beta_i}$ -module (cf. [5, 5.1.4]).

4.4. Type B

We shall check that the weight $w^{\chi^{(\alpha,\beta)}}$ given by formula (4.2.1) is equal to the $w_{q,q;z,z}^{(\alpha,\beta)}$ with $z = (q-1)r/(1+r)$ determined in Section 3. By Remark 2.9, we have

$$\left(\frac{1-q}{1+r}\right)^n P_{\chi^{(\alpha,\beta)}}(q,r) = \frac{\text{Fdeg}_{\chi^{(\alpha,\beta)}}}{(1+r)^n P_{W_n}} \prod_{x \in \alpha} (1+q^{2c(x)+1}r) \prod_{x \in \beta} (1+q^{2c(x)-1}r).$$

On the other hand, by Remark 3.6, we have

$$\begin{aligned} w_{q,q;z,z}^{(\alpha,\beta)} \Big|_{z=\frac{(q-1)r}{1+r}} &= \frac{\text{Gdeg}_{\chi^{(\alpha,\beta)}}}{(q-1)^n P_{W_n}} \prod_{x \in \alpha} \left(q-1 + (q^{2c(x)+1} - 1) \frac{(q-1)r}{1+r} \right) \\ &\quad \times \prod_{x \in \beta} \left(q-1 + (q^{2c(x)-1} - 1) \frac{(q-1)r}{1+r} \right) \\ &= \frac{\text{Gdeg}_{\chi^{(\alpha,\beta)}}}{(1+r)^n P_{W_n}} \prod_{x \in \alpha} (1 + q^{2c(x)+1} r) \prod_{x \in \beta} (1 + q^{2c(x)-1} r). \end{aligned}$$

By (4.1.1), it is sufficient to show that

$$\prod_{x \in \alpha} (1 + q^{2c(x)+1} r) \prod_{x \in \beta} (1 + q^{2c(x)-1} r) \quad (4.4.1)$$

has constant value on each family, which has been proved in [6]. In fact, let

$$\begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_{m+1} \\ \mu_1 & \mu_2 & \cdots & \mu_m \end{pmatrix} \quad \left(\text{respectively } \begin{pmatrix} \lambda'_1 & \lambda'_2 & \cdots & \lambda'_{m+1} \\ \mu'_1 & \mu'_2 & \cdots & \mu'_m \end{pmatrix} \right)$$

be a symbol of $\chi^{(\alpha,\beta)}$ (respectively $\chi^{(\alpha',\beta')}$). Two characters $\chi^{(\alpha,\beta)}$ and $\chi^{(\alpha',\beta')}$ of W_n lie in the same family if and only if the unordered sets of entries of their symbols are the same including multiplicities. By [14, I.1.Ex.3], it is easily seen that (4.4.1) has constant value on each family. This completes the proof.

4.5. Type D

In this case, the situation is different from the case of type B_n . In the case of type B_n , the value of (4.4.1) is constant on each family. But in this case, this property is not satisfied. For example, in the case of type D_4 , irreducible characters $\chi^{\{(2 \cdot 1), (1)\}}$, $\chi^{\{(2^2), \cdot\}}$ and $\chi^{\{(1^2), (2)\}}$ are in the same family, and by Remark 2.11 we have a following.

Character	Symbol	$(\text{Fdeg}_{\chi^{(\alpha,\beta)}})^{-1} (1-q)^n P_{\tilde{W}_4} P_{\chi^{(\alpha,\beta)}}(q, r)$
$\chi^{\{(2 \cdot 1), (1)\}}$	$\begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$	$(1 + q^{-1}r)(1 + qr)(1 + 2q^{-1}(1 + q^2)^{-1}(1 + q^4)r + r^2)$
$\chi^{\{(2^2), \cdot\}}$	$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$	$(1 + q^{-1}r)(1 + qr)(1 + 2q(1 + q^2)(1 + q^4)^{-1}r + r^2)$
$\chi^{\{(1^2), (2)\}}$	$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$	$(1 + q^{-1}r)(1 + qr)(1 + 2^{-1}q^{-3}(1 + q^2)(1 + q^4)r + r^2)$

So we should calculate w^χ given by (4.2.1) explicitly. Before we start to prove the theorem, we prepare a proposition. We use the same argument as in [10, §1].

Let Y be a set with $2s$ elements and $\mathfrak{P}$ (respectively $\mathfrak{P}_i$) the set of all subsets of Y (respectively the set of all subsets $I \subset Y$ with $|I| \equiv i \pmod{2}$ for a given integer i). Let K

be a field of characteristic 0. For a function $f: \mathfrak{P} \rightarrow K$, we define the Fourier transform $\hat{f}: \mathfrak{P} \rightarrow K$ by

$$\hat{f}(I) = 2^{-s} \sum_{J \in \mathfrak{P}} (-1)^{|I \cap J|} f(J).$$

It is easily checked that $\hat{\hat{f}} = f$. We fix a function $\zeta: \mathfrak{P} \rightarrow K$. Let $\bar{y}$ be an element not contained in Y and put $\bar{I} = I \cup \{\bar{y}\}$ for $I \subset Y$. Assume that $\bar{Y}$ has a given total order such that $\bar{y}$ is the minimal element. We define $f_{\pm}: \mathfrak{P} \rightarrow K$ and $\varphi: \mathfrak{P} \rightarrow K$ by

$$\begin{aligned} f_{\pm}(I) &= 2^{-s} \prod_{y \in I} (\zeta(y) \mp \zeta(\bar{y})) \prod_{y \in I'} (\zeta(y) \pm \zeta(\bar{y})) \\ &\quad \times \prod_{\substack{(y, y') \in I \times I \\ y > y'}} (\zeta(y) - \zeta(y')) \prod_{\substack{(y, y') \in I' \times I' \\ y > y'}} (\zeta(y) - \zeta(y')) \\ &\quad \times \prod_{(y, y') \in I \times I'} (\zeta(y) + \zeta(y')), \quad \text{if } I \in \mathfrak{P}_s, \end{aligned}$$

and $f_{\pm}(I) = 0$ if $I \notin \mathfrak{P}_s$, and

$$\begin{aligned} \varphi(I) &= \left(\prod_{y \in I'} \zeta(y) \prod_{y \in I} (\zeta(y)^2 - \zeta(\bar{y})^2) + \prod_{y \in I} \zeta(y) \prod_{y \in I'} (\zeta(y)^2 - \zeta(\bar{y})^2) \right) \\ &\quad \times \prod_{\substack{(y, y') \in I \times I \\ y > y'}} (\zeta(y)^2 - \zeta(y')^2) \prod_{\substack{(y, y') \in I' \times I' \\ y > y'}} (\zeta(y)^2 - \zeta(y')^2), \quad \text{if } |I| = s, \end{aligned}$$

and $\varphi(I) = 0$ if $|I| \neq s$ where $I' = Y - I$. Then we have:

Proposition 4.4. *The functions on $\mathfrak{P}$*

$$\begin{aligned} I &\mapsto (-1)^{c(I) + \binom{|I|+1}{2}} (f_+(I) + f_-(I)) \quad \text{and} \\ I &\mapsto (-1)^{c(I) + \binom{|I|}{2}} \varphi(I) \end{aligned}$$

correspond to each other under the Fourier transform. Here $c(I) = \sharp\{(y, y') \in I \times I' \mid y < y'\}$.

Proof. For $J \subset \bar{Y}$, let v_J be the characteristic function of J and $\delta_J(y) = (-1)^{v_J(y)}$ for $y \in \bar{Y}$. For $I \in \mathfrak{P}_s$, $f'_{\pm}(I) = (-1)^{c(I) + \binom{|I|+1}{2}} f_{\pm}(I)$ is given by

$$\begin{aligned}
 f'_+(I) &= (-1)^{|I|} 2^{-s} \prod_{\substack{(y,y') \in \bar{Y} \times \bar{Y} \\ y > y'}} (\delta_{\bar{I}}(y)\zeta(y) - \delta_{\bar{I}}(y')\zeta(y')) \\
 &= (-1)^{|I|} 2^{-s} \sum_{\sigma} \epsilon(\sigma) \prod_{y \in \bar{Y}} (\delta_{\bar{I}}(y)\zeta(y))^{\sigma(y)}
 \end{aligned}$$

and

$$\begin{aligned}
 f'_-(I) &= (-1)^{|I|} 2^{-s} \prod_{\substack{(y,y') \in \bar{Y} \times \bar{Y} \\ y > y'}} (\delta_I(y)\zeta(y) - \delta_I(y')\zeta(y')) \\
 &= (-1)^{|I|} 2^{-s} \sum_{\sigma} \epsilon(\sigma) \prod_{y \in \bar{Y}} (\delta_I(y)\zeta(y))^{\sigma(y)},
 \end{aligned}$$

where σ runs over all bijections $\bar{Y} \rightarrow \{0, 1, \dots, 2s\}$ and $\epsilon(\sigma) = 1$ or -1 according as $\sharp\{(y, y') \in \bar{Y} \times \bar{Y} \mid y > y', \sigma(y) < \sigma(y')\}$ is even or odd. On the other hand, for $I \subset Y$ with $|I| = s$, $\varphi'(I) = (-1)^{c(I) + \binom{|I|}{2}} \varphi(I)$ is given by

$$\begin{aligned}
 \varphi'(I) &= \left(\sum_{\sigma_1} \epsilon(\sigma_1) \prod_{y \in \bar{I}} \zeta(y)^{\sigma_1(y)} \right) \left(\sum_{\sigma'_2} \epsilon(\sigma'_2) \prod_{y \in I'} \zeta(y)^{\sigma'_2(y)} \right) \\
 &\quad + \left(\sum_{\sigma'_1} \epsilon(\sigma'_1) \prod_{y \in \bar{I}'} \zeta(y)^{\sigma'_1(y)} \right) \left(\sum_{\sigma_2} \epsilon(\sigma_2) \prod_{y \in I} \zeta(y)^{\sigma_2(y)} \right),
 \end{aligned}$$

where σ_1 (respectively σ'_1) runs over all bijections from $\bar{I}$ (respectively $\bar{I}'$) to $\{0, 2, \dots, 2s\}$, σ_2 (respectively σ'_2) runs over all bijections from I (respectively I') to $\{1, 3, \dots, 2s-1\}$ and ϵ is defined as above. Applying the Fourier transform to $f'_{\pm}$, we have

$$\hat{f}'_+(I) = (-1)^s 2^{-2s} \sum_{\sigma} \epsilon(\sigma) (-1)^{\sigma(\bar{y})} \prod_{y \in \bar{Y}} \zeta(y)^{\sigma(y)} \sum_{J \in \mathfrak{P}_s} \prod_{y \in Y} \delta_J(y)^{\nu_I(y) + \sigma(y)}$$

and

$$\hat{f}'_-(I) = (-1)^s 2^{-2s} \sum_{\sigma} \epsilon(\sigma) \prod_{y \in \bar{Y}} \zeta(y)^{\sigma(y)} \sum_{J \in \mathfrak{P}_s} \prod_{y \in Y} \delta_J(y)^{\nu_I(y) + \sigma(y)}.$$

Note that

$$\begin{aligned}
 &\sum_{J \in \mathfrak{P}_s} \prod_{y \in Y} \delta_J(y)^{\nu_I(y) + \sigma(y)} \\
 &= \begin{cases} 2^{2s-1}, & \text{if } \sigma(I) = \{1, 3, \dots, 2s-1\}, & (*)1 \\ (-1)^s 2^{2s-1}, & \text{if } \sigma(\bar{I}) = \{0, 2, \dots, 2s\}, & (*)2 \\ 2^{2s-1}, & \text{if } \sigma(\bar{I}) = \{1, 3, \dots, 2s-1\}, & (*)3 \\ (-1)^s 2^{2s-1}, & \text{if } \sigma(I) = \{0, 2, \dots, 2s\}, & (*)4 \\ 0, & \text{o.w.} \end{cases} \quad (4.5.1)
 \end{aligned}$$

If $|I| = s$, then by (4.5.1) we can easily see that

$$\hat{f}'_+(I) + \hat{f}'_-(I) = (-1)^s \sum_{\sigma: (*1)} \epsilon(\sigma) \prod_{y \in \bar{Y}} \zeta(y)^{\sigma(y)} + \sum_{\sigma: (*2)} \epsilon(\sigma) \prod_{y \in \bar{Y}} \zeta(y)^{\sigma(y)} = \varphi'(I),$$

as required. If $|I| = s \pm 1$, then it is enough to consider the cases $(*3)$ and $(*4)$ in (4.5.1). In both cases, $\sigma(\bar{y})$ is odd and we can easily see that $\hat{f}'_+(I) + \hat{f}'_-(I) = 0$. If $|I| \neq s$ nor $s \pm 1$, then it is obvious from (4.5.1) that $\hat{f}'_+(I) = \hat{f}'_-(I) = 0$. This completes the proof. $\square$

Now we shall show that the trace function τ given in Theorem 4.1 coincides with $\tau_{q;z}^{D_n}$ determined in Section 3 where $z = (q-1)r/(1+r)$. That is, we shall show that the $w^{\chi^{(\alpha,\beta)}}$ (respectively $w^{\chi^{(\alpha,\pm)}}$) given by (4.2.1) satisfy

$$w^{\chi^{(\alpha,\beta)}} = w_{q;z}^{\{\alpha,\beta\}} \Big|_{z=\frac{(q-1)r}{1+r}} \quad (\text{respectively } w^{\chi^{(\alpha,\pm)}} = w_{q;z}^{\{\alpha,\pm\}} \Big|_{z=\frac{(q-1)r}{1+r}}), \quad (4.5.2)$$

where the $w_{q;z}^{\{\alpha,\beta\}}$ and the $w_{q;z}^{\{\alpha,\pm\}}$ are the weights of $\tau_{q;z}^{D_n}$. First we prove (4.5.2) for $\chi^{(\alpha,\pm)}$. The character $\chi^{(\alpha,\pm)}$ forms a one-element family and its fake degree and generic degree are equal. By Remark 2.11, we have

$$\begin{aligned} w^{\chi^{(\alpha,\pm)}} &= \left(\frac{1-q}{1+r} \right)^n P_{\chi^{(\alpha,\pm)}}(q, r) \\ &= \frac{\text{Fdeg}_{\chi^{(\alpha,\pm)}}}{(1+r)^n P_{\tilde{W}_n}} \prod_{x \in \alpha} (1 + rq^{2c(x)+1})(1 + rq^{2c(x)-1}). \end{aligned}$$

On the other hand, by Theorem 3.10, we have

$$\begin{aligned} w_{q;z}^{\{\alpha,\pm\}} \Big|_{z=\frac{(q-1)r}{1+r}} &= \frac{\text{Gdeg}_{\chi^{(\alpha,\pm)}}}{(q-1)^n P_{\tilde{W}_n}} \prod_{x \in \alpha} \left((q-1) + (q^{2c(x)+1} - 1) \frac{(q-1)r}{1+r} \right) \\ &\quad \times \left((q-1) + (q^{2c(x)-1} - 1) \frac{(q-1)r}{1+r} \right) \\ &= \frac{\text{Gdeg}_{\chi^{(\alpha,\pm)}}}{(1+r)^n P_{\tilde{W}_n}} \prod_{x \in \alpha} (1 + rq^{2c(x)+1})(1 + rq^{2c(x)-1}). \end{aligned}$$

By (4.1.1), we have

$$w^{\chi^{(\alpha,\pm)}} = w_{q;z}^{\{\alpha,\pm\}} \Big|_{z=\frac{(q-1)r}{1+r}},$$

as required.

Next we prove (4.5.2) for $w^{\chi^{(\alpha,\beta)}}$. If we multiply both sides of (4.5.2) by $(1+r)^n$, then they are contained in $\mathbb{Q}[[q]][r]$, that is, they are polynomials with respect to “ r ”. Thus it is

sufficient to show that the equation is satisfied for infinitely many specialized r . Now we specialize $r \mapsto -q^{2m+1}$ for some large $m (\geq n)$. Let

$$\begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_m \\ \mu_1 & \mu_2 & \cdots & \mu_m \end{pmatrix}$$

be a symbol of $\chi^{\{\alpha, \beta\}}$, that is, $\lambda_i = \alpha_i + i - 1$ ($1 \leq i \leq m$) and $\mu_j = \beta_j + j - 1$ ($1 \leq j \leq m$). By Theorem 2.10, we have

$$\begin{aligned} P_{\chi^{\{\alpha, \beta\}}}(q, -q^{2m+1}) &= \frac{q^{2n(\alpha) + 2n(\beta)}}{\prod_{x \in \alpha} (1 - q^{2h(x)}) \prod_{x \in \beta} (1 - q^{2h(x)})} \\ &\quad \times \left(q^{|\beta|} \prod_{x \in \alpha} (1 - q^{2(c(x) + m + 1)}) \prod_{x \in \beta} (1 - q^{2(c(x) + m)}) \right. \\ &\quad \left. + q^{|\alpha|} \prod_{x \in \alpha} (1 - q^{2(c(x) + m)}) \prod_{x \in \beta} (1 - q^{2(c(x) + m + 1)}) \right). \end{aligned}$$

By [14, I.1.Ex.1 and I.1.Ex.3], we have

$$\begin{aligned} P_{\chi^{\{\alpha, \beta\}}}(q, -q^{2m+1}) &= \frac{(q^{|\mu|} \prod_{i=1}^m (q^{2(\lambda_i + 1)} - 1) + q^{|\lambda|} \prod_{i=1}^m (q^{2(\mu_i + 1)} - 1)) \prod_{1 \leq i' < i \leq m} (q^{2\lambda_i} - q^{2\lambda_{i'}}) \prod_{1 \leq i' < i \leq m} (q^{2\mu_i} - q^{2\mu_{i'}})}{q^{\binom{2m-2}{2} + \binom{2m-4}{2} + \cdots} \prod_{i=1}^m \prod_{k=1}^{m-i} (q^{2k} - 1) \prod_{i=1}^m \prod_{k=1}^{m-i+1} (q^{2k} - 1)}. \end{aligned}$$

Since $z = (q - 1)r / (1 + r)$, z specializes to $q^{2m+1}(q - 1) / (q^{2m+1} - 1)$ through $r \mapsto -q^{2m+1}$. By Theorem 3.10 and [loc. cit.], we have

$$\begin{aligned} w_{q; z}^{\{\alpha, \beta\}} \Big|_{z = \frac{q^{2m+1}(q-1)}{q^{2m+1}-1}} &= D_q^{\{\alpha, \beta\}} \times \left(R_{1, q}^{(\alpha, \beta)} \left(\frac{q^m(q-1)}{q^{2m+1}-1}, \frac{q^{2m+1}(q-1)}{q^{2m+1}-1} \right) \right. \\ &\quad \left. + R_{1, q}^{(\beta, \alpha)} \left(\frac{q^m(q-1)}{q^{2m+1}-1}, \frac{q^{2m+1}(q-1)}{q^{2m+1}-1} \right) \right) \\ &= \left(\frac{q-1}{q^{2m+1}-1} \right)^n \times \left(\prod_{i=1}^m (q^{\lambda_i+1} - 1)(q^{\mu_i+1} + 1) + \prod_{i=1}^m (q^{\lambda_i+1} + 1)(q^{\mu_i+1} - 1) \right) \\ &\quad \times \frac{\prod_{1 \leq i' < i \leq m} (q^{\lambda_i} - q^{\lambda_{i'}}) \prod_{1 \leq i' < i \leq m} (q^{\mu_i} - q^{\mu_{i'}}) \prod_{1 \leq i, i' \leq m} (q^{\lambda_i} + q^{\mu_{i'}})}{2^m q^{\binom{2m-2}{2} + \binom{2m-4}{2} + \cdots} \prod_{i=1}^m \prod_{k=1}^{m-i} (q^{2k} - 1) \prod_{i=1}^m \prod_{k=1}^{m-i+1} (q^{2k} - 1)}. \end{aligned}$$

We assume that $\chi^{\{\alpha,\beta\}}$ and $\chi^{\{\alpha',\beta'\}}$ lie in the same family. Then the unordered sets of entries of their symbols

$$\begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_m \\ \mu_1 & \mu_2 & \cdots & \mu_m \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \lambda'_1 & \lambda'_2 & \cdots & \lambda'_m \\ \mu'_1 & \mu'_2 & \cdots & \mu'_m \end{pmatrix}$$

are the same including multiplicities. Let Y be the set of numbers which appear just once in $\{\lambda_1, \lambda_2, \dots, \lambda_m, \mu_1, \mu_2, \dots, \mu_m\}$, and put $I = Y \cap \{\lambda_1, \lambda_2, \dots, \lambda_m\}$ and $J = Y \cap \{\lambda'_1, \lambda'_2, \dots, \lambda'_m\}$. Then the entry of the Fourier transform matrix corresponding to $(\chi^{\{\alpha,\beta\}}, \chi^{\{\alpha',\beta'\}})$ is given by

$$2^{-\frac{|Y|}{2}+1}(-1)^{|I \cap J|+c(I)+c(J)+\binom{|I|+1}{2}+\binom{|J|}{2}} = 2^{-\frac{|Y|}{2}+1}(-1)^{|I^\# \cap J^\#|},$$

where $c(I) = \sharp\{(y, y') \in I \times (Y - I) \mid y < y'\}$ and $I^\# = (I \cup Z) - (I \cap Z)$ with $Z = \{y_1, y_3, \dots, y_{2s-1}\} \subset Y = \{y_1 < y_2 < y_3 < \dots < y_{2s}\}$ (cf. [10,11]). Applying Proposition 4.4 with $\zeta(y) = q^y$ and $\bar{y} = -1$, we can complete the proof.

4.6. Type $I_2(m)$

Let W be the Coxeter group of type $I_2(m)$ and $\mathcal{H}$ the corresponding Hecke algebra with generators $\{T_1, T_2\}$ and a parameter q . We use the notation in Section 2.5. The set of irreducible characters $\text{Irr } W$ is decomposed into two singletons $\{\phi_{1,0}\}$, $\{\phi_{1,m}\}$ and a family $\mathcal{F}$ which consists of the remaining ones. By [13], the entries of the Fourier transform matrix of $\mathcal{F}$ are given as follows:

$$\begin{aligned} \{\phi_{2,j}, \phi_{2,k}\} &= \{\phi_{2,k}, \phi_{2,j}\} = \frac{2 - \xi^{jk} - \xi^{-jk}}{m}, \quad \text{for } 1 \leq j, k \leq \left[\frac{m-1}{2}\right], \\ \{\phi_{2,j}, \phi'_{1,\frac{m}{2}}\} &= \{\phi'_{1,\frac{m}{2}}, \phi_{2,j}\} = \{\phi_{2,j}, \phi''_{1,\frac{m}{2}}\} = \{\phi''_{1,\frac{m}{2}}, \phi_{2,j}\} \\ &= \frac{1 - (-1)^j}{m}, \quad \text{for } 1 \leq j \leq \left[\frac{m-1}{2}\right], \\ \{\phi'_{1,\frac{m}{2}}, \phi'_{1,\frac{m}{2}}\} &= \{\phi''_{1,\frac{m}{2}}, \phi''_{1,\frac{m}{2}}\} = \frac{1 - (-1)^{\frac{m}{2}} + m}{m}, \\ \{\phi'_{1,\frac{m}{2}}, \phi''_{1,\frac{m}{2}}\} &= \{\phi''_{1,\frac{m}{2}}, \phi'_{1,\frac{m}{2}}\} = \frac{1 - (-1)^{\frac{m}{2}} - m}{m}, \end{aligned}$$

where $\xi = e^{2\pi\sqrt{-1}/m}$. Let $w_{i,j}$ (respectively $w'_{1,m/2}$, $w''_{1,m/2}$) be the weight corresponding to $\phi_{i,j}$ (respectively $\phi'_{1,m/2}$, $\phi''_{1,m/2}$) determined by Eq. (4.2.1). Direct calculation shows that

$$w_{1,0} = \frac{(z+1)((q^{m-1}-1)z+q-1)}{(q+1)(q^m-1)},$$

$$\begin{aligned}
w_{1,m} &= \frac{(z-q)((q^{m-1}-1)z - q^{m-1}(q-1))}{(q+1)(q^m-1)}, \\
w_{2,j} &= \frac{(2-\xi^j-\xi^{-j})(z+1)(q-z)}{m(q-\xi^j)(q-\xi^{-j})}, \quad \text{for } 1 \leq j \leq \left[\frac{m-1}{2}\right], \\
w'_{1,\frac{m}{2}} &= w''_{1,\frac{m}{2}} = \frac{2(z+1)(q-z)}{m(q+1)^2}.
\end{aligned}$$

Let τ be the trace function on $\mathcal{H}$ determined by these weights. In [9], Kihara showed the next theorem.

Theorem 4.5. τ satisfies the following equations.

- (1) $\tau(1) = 1$.
- (2) $\tau(T_1) = \tau(T_2) = z$.
- (3) $\tau(\underbrace{T_1 T_2 \cdots T_{i+1}}_{i+1} \cdots \underbrace{T_1^{-1} T_2^{-1}}_{i-1}) = z^2$, for $1 \leq i \leq \left[\frac{m-1}{2}\right]$.

In particular τ satisfies the Markov property.

Remark 4.6. The equations in the theorem determine the trace function on $\mathcal{H}$ uniquely. For even m , let $\mathcal{H}_{Q,q}$ be the Iwahori–Hecke algebra of type $I_2(m)$ with two parameters Q and q . For two indeterminates y and z , we can also construct a trace function τ on $\mathcal{H}_{Q,q}$ which satisfies

- (1) $\tau(1) = 1$.
- (2) $\tau(T_1) = y$.
- (3) $\tau(T_2) = z$.
- (4) $\tau(\underbrace{T_1 T_2 \cdots T_{i+1}}_{i+1} \cdots \underbrace{T_1^{-1} T_2^{-1}}_{i-1}) = \begin{cases} yz, & \text{if } i \text{ is odd,} \\ y^2, & \text{if } i \text{ is even,} \end{cases} \quad \text{for } 1 \leq i \leq \left[\frac{m-1}{2}\right]$.

Kihara has determined the weights of τ in [9]:

$$\begin{aligned}
w_{1,0} &= \frac{(y+1)((1-q^{\frac{m}{2}}Q^{\frac{m}{2}})z + (1-q^{\frac{m}{2}-1}Q^{\frac{m}{2}-1})qy - qQ + 1)}{(Q+1)(q+1)(1-q^{\frac{m}{2}}Q^{\frac{m}{2}})}, \\
w_{1,m} &= \frac{(y-Q)((1-q^{\frac{m}{2}}Q^{\frac{m}{2}})z + (1-q^{\frac{m}{2}-1}Q^{\frac{m}{2}-1})qy + q^{\frac{m}{2}+1}Q^{\frac{m}{2}} - q^{\frac{m}{2}}Q^{\frac{m}{2}-1})}{(Q+1)(q+1)(1-q^{\frac{m}{2}}Q^{\frac{m}{2}})}, \\
w'_{1,\frac{m}{2}} &= \frac{(y+1)(-(q^{\frac{m}{2}} - Q^{\frac{m}{2}})z + (q^{\frac{m}{2}-1} - Q^{\frac{m}{2}-1})qy + q^{\frac{m}{2}+1} - q^{\frac{m}{2}}Q)}{(Q+1)(q+1)(q^{\frac{m}{2}} - Q^{\frac{m}{2}})}, \\
w''_{1,\frac{m}{2}} &= \frac{(y-Q)(-(q^{\frac{m}{2}} - Q^{\frac{m}{2}})z + (q^{\frac{m}{2}-1} - Q^{\frac{m}{2}-1})qy + Q^{\frac{m}{2}} - qQ^{\frac{m}{2}-1})}{(Q+1)(q+1)(q^{\frac{m}{2}} - Q^{\frac{m}{2}})},
\end{aligned}$$

$$w_{2,j} = \frac{q(4 - (\zeta^j + \zeta^{-j})^2)(1+y)(Q-y)}{m(q + (\zeta^j + \zeta^{-j})\sqrt{qQ} + Q)(qQ - (\zeta^j + \zeta^{-j})\sqrt{qQ} + 1)},$$

for $j = 1, 2, \dots, \frac{m}{2} - 1$.

4.7. Remark. Thus we proved the main Theorem 4.1 by direct calculation. I believe that there should exist a hidden geometrical meaning for the theorem and that it gives us a proof which does not depend on a case-by-case argument.

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